DAEN 300: Data Engineering Coding Experience I

Industrial and Systems Engineering, Texas A&M University

Fall 2027

Course Information

- **Credit Hours:** 1, **Contact Hours:** 0 lecture hours, 3 laboratory hours per week
- **Prerequisites:** Grade of C or better in ECEN 360; concurrent enrollment in DAEN 321 and DAEN 331; junior or senior classification.
- **Format:** Hands-on computational laboratory, **Programming Language:** Python
- **Instructor:** Alexander Abuabara, **Office hours:** Wednesdays 1-3 PM, ETB 4013
  `https://engineering.tamu.edu/industrial/profiles/abuabara-alexander.html`

Course Description

Application of computational tools to model and solve data engineering problems primarily involving machine learning and optimization techniques.

This laboratory course develops practical programming skills for solving data engineering problems. Students prepare and analyze data, implement machine learning models, apply optimization techniques such as loss minimization and gradient descent, and solve basic constrained optimization problems. Activities emphasize interpretation of results and reinforce concepts from concurrent courses.

Course Learning Objectives

Upon successful completion of this course, students will be able to:

1. Develop Python programs to process, analyze, and visualize engineering datasets.
2. Implement and evaluate basic machine learning models for prediction and classification.
3. Formulate and solve constrained optimization problems using computational tools.
4. Interpret computational results, assess model performance, and communicate findings.
5. Develop reproducible and documented computational solutions, including collaborative workflows.

Course Topics

1. **Computational Foundations**
   - Python and Jupyter notebooks
   - Numerical computing with NumPy
   - Data manipulation with pandas
   - Data visualization
   - Code documentation and version control
2. **Machine Learning**
   - Regression
   - Classification
   - Training and testing datasets
   - Model evaluation and validation

- Engineering prediction applications

3. **Optimization**
   - Objective functions and constraints
   - Linear programming
   - Numerical optimization
   - Engineering resource allocation problems
4. **Integrated Applications**
   - Combining prediction and optimization
   - Reproducible computational workflows
   - Interpretation and communication of results

Tentative Course Schedule

Week	Laboratory Topic
1	Python environment, Jupyter notebooks, and Git
2	Numerical computing with NumPy
3	Data manipulation and cleaning with pandas
4	Exploratory data analysis and visualization
5	Linear regression
6	Classification
7	Model validation and performance evaluation
8	Machine learning case study
9	Optimization formulation and constraints
10	Linear programming with SciPy
11	Constrained numerical optimization
12	Engineering optimization case study
13	Integrated project development
14	Integrated project implementation and testing
15	Project demonstration and presentation

Software and Computational Tools

The course uses freely available computational tools:

- Python 3, JupyterLab or Google Colab
- NumPy and pandas
- Matplotlib
- scikit-learn
- SciPy optimization tools
- Git and GitHub

Course Materials

Primary Reference:

- *Data Science: A First Introduction (Python Edition).* University of British Columbia. `https://github.com/UBC-DSCI/introduction-to-datascience-python`

Supplementary Reference:

- *Mathematical Programming and Operations Research.*
  `https://github.com/open-optimization/open-optimization-or-book`
- Additional Python notebooks, datasets, and laboratory instructions will be provided by the instructor.

Assessment and Grading

Assessment Component	Weight
Weekly coding laboratories	60%
Machine learning case study	15%
Integrated final project	20%
Participation and code documentation	5%
Total	**100%**

Assignments will be evaluated based on correctness, appropriate use of computational methods, interpretation of results, and code clarity and reproducibility.

Integrated Final Project

Students will complete a small computational project combining machine learning and optimization.

The project should include:

1. Data preparation and exploratory analysis.
2. Development and evaluation of a predictive model.
3. Formulation of an optimization problem.
4. Computational solution and interpretation.
5. Submission of a documented Jupyter notebook and a brief summary of findings.

Possible applications include demand forecasting, manufacturing planning, and resource allocation.

Instructional Approach

The course emphasizes learning through hands-on coding. Laboratory meetings will consist of short demonstrations, guided exercises, and independent or collaborative programming activities.

Students are expected to apply mathematical and engineering concepts introduced in prerequisite and concurrent courses.

Course Expectations

Students are expected to:

- Attend and participate in laboratory sessions.
- Complete and submit coding assignments on time.
- Write readable and appropriately documented code.
- Interpret and explain computational results.
- Follow university academic integrity policies.
- Acknowledge external code, libraries, and AI-assisted contributions as required by the instructor.